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The effects of multiregional fascial mobilization on heart rate and physiological parameters in patients with obstructive sleep apnea

A randomized controlled trial

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Abstract

Background: Obstructive sleep apnea (OSA) is associated with increased sympathetic activity and an elevated cardiovascular risk. Noninvasive interventions targeting autonomic regulation may offer complementary therapeutic strategies.

Methods: This randomized controlled trial aimed to investigate the acute effects of multiregional fascial mobilization on the physiological parameters of patients with OSA. Patients with polysomnography-confirmed moderate-to-severe OSA (apnea-hypopnea index ≥ 15 events/hour) were randomly assigned to an intervention group that received a standardized 16-minute multiregional fascial mobilization protocol or a control group that rested quietly in the supine position for the corresponding period without receiving manual or tactile intervention. The intervention targeted the cervical, thoracic, and abdominal regions, under controlled conditions. The primary outcomes included heart rate, systolic and diastolic blood pressure, and peripheral oxygen saturation. Measurements were obtained at baseline, immediately post-intervention, and 5 and 15 minutes post-intervention.

Results: A total of 95 male patients (mean age: 48.12 ± 10.21 years) completed the study. Baseline characteristics were comparable between the groups, except for heart rate. After adjustment for baseline heart rate using mixed analysis of covariance, no significant group × time interaction was observed for heart rate ($P = .476$, partial $\eta p^2 = .008$). Similarly, no significant group × time interactions were observed for systolic blood pressure, diastolic blood pressure, or peripheral oxygen saturation (all $P > .05$). Although significant time effects were observed for some physiological parameters, the magnitude of these effects varied across outcomes, and none was accompanied by a significant group × time interaction. Therefore, these temporal changes were not interpreted as intervention-specific effects. No adverse events were reported.

Conclusion: Heart rate changed significantly over time following the interventions; however, after adjustment for baseline heart rate, the pattern of change did not differ significantly between the fascial mobilization and control groups. Similarly, no significant between-group differences in changes over time were observed for blood pressure or oxygen saturation. These findings do not demonstrate a specific acute effect of multiregional fascial mobilization on the physiological parameters assessed. Further studies using advanced measures of autonomic function and longer follow-up periods are warranted.

Abbreviations: ANCOVA = analysis of covariance, HRV = heart rate variability, OSA = obstructive sleep apnea, PSG = polysomnography, RCT = randomized controlled trial, SpO_2 = peripheral oxygen saturation.

Keywords: autonomic nervous system, fascial mobilization, heart rate, obstructive sleep apnea, randomized controlled trial

The authors have no funding and conflicts of interest to declare.

The datasets generated during and/or analyzed during the current study are not publicly available, but are available from the corresponding author on reasonable request.

This randomized controlled trial was conducted at the Sleep Laboratory of Mardin Training and Research Hospital. The study was registered on ClinicalTrials.gov (NCT07169058) and was conducted in accordance with the principles of the Declaration of Helsinki and Good Clinical Practice guidelines. Ethical approval was obtained from the nonInvasive Clinical Research Ethics Committee of Mardin Artuklu University (Decision No: 2025/2-11; Date: February 18, 2025). Institutional permission to conduct the study at Mardin Training and Research Hospital was obtained from the Mardin Provincial Health Directorate (Permission No:266215234; Date: January 23, 2025). All participants provided written informed consent prior to enrollment and voluntarily participated without financial compensation.

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1. Introduction

Obstructive sleep apnea (OSA) is a prevalent sleep-related breathing disorder characterized by repeated episodes of partial or complete upper airway obstruction during sleep.[1] The respiratory restriction that occurs during these episodes affects autonomic nervous system functions in both the acute and chronic phases, and it has been reported that sympathetic activity is markedly increased.[2] The autonomic response to physiological stress stimuli plays a critical role in maintaining homeostasis, particularly through the circulatory and respiratory systems.[3] Cardiovascular adaptations are generally achieved through the reciprocal interaction of the sympathetic and parasympathetic systems, regulating oxygen delivery to tissues and perfusion of vital organs.[4] In OSA, these responses are mostly triggered by hypoxia and apnea events. During hypoxia, the activation of chemoreceptors stimulates hyperventilation, while simultaneous sympathetic-mediated vasoconstriction directs oxygen toward vital organs. Subsequently, bradycardia mediated by the parasympathetic system aims to preserve balance by reducing myocardial oxygen consumption.[4,5] This pattern resembles evolutionary mechanisms defined as the diving reflex or oxygen-conserving reflex.[6] However, in OSA, chronic repetitive hypoxia, accompanied by sustained sympathetic overactivity, imposes a pathological burden on the cardiovascular system.[3–5]

Indeed, arousal and hypoxemia resulting from apnea and hypopnea episodes are associated with increased sympathetic activity and adverse cardiovascular and vascular consequences.[7,8] The literature reports that as the severity of OSA increases, reductions in heart rate reserve, chronotropic dysfunction, elevated systolic and diastolic blood pressure values, and cardiac arrhythmias are observed.[9–12] In addition, it has been demonstrated that even at rest, individuals with OSA exhibit higher arterial blood pressure values compared to healthy controls.[13]

The cranial nerve X, the vagus nerve, is an important component of parasympathetic regulation and contributes to the regulation of cardiac, respiratory, gastrointestinal, and inflammatory processes.[14–16] Invasive and noninvasive forms of direct vagus nerve stimulation have therefore been investigated in several clinical contexts.[17,18] However, these approaches are mechanistically distinct from manual therapy, and anatomical proximity of a manual intervention to the course of the vagus nerve does not establish vagal stimulation. Evidence regarding the acute physiological effects of fascial and other manual interventions remains limited and heterogeneous, particularly in individuals with OSA. Accordingly, heart rate, blood pressure, and peripheral oxygen saturation (SpO_2) may be used to characterize short-term physiological responses, but they cannot serve as specific measures of vagal activation.

This randomized controlled study was conducted to compare the acute physiological responses to a prespecified multiregional fascial mobilization protocol with those observed during a time-matched resting control condition in individuals with OSA. The intervention involved cervical, thoracic, and abdominal regions selected according to anatomical and fascial considerations. Heart rate, systolic and diastolic blood pressure, and SpO_2 were assessed to determine whether their short-term patterns of change differed between the fascial mobilization and control groups. We hypothesized that the temporal responses in these physiological parameters would differ between the 2 groups.

2. Materials and methods

2.1. Study design

A parallel-group randomized controlled trial (RCT) was conducted at the Sleep Laboratory of Mardin Training and Research Hospital. The study was registered on ClinicalTrials.gov (NCT07169058) and was reported in accordance with the consolidated standards of reporting trials recommendations. Ethical approval was obtained from the Noninvasive Clinical Research Ethics Committee of Mardin Artuklu University (Decision No: 2025/2-11; Date: February 18, 2025). Institutional permission to conduct the study at Mardin Training and Research Hospital was obtained from the Mardin Provincial Health Directorate (Permission No:266215234; Date: January 23, 2025).

Subjects were recruited from patients referred to the sleep laboratory for polysomnography (PSG) evaluation. Recruitment, screening, and data collection were performed by a multidisciplinary team consisting of pulmonologists and physiotherapists according to predefined inclusion and exclusion criteria.

The inclusion criteria were as follows: patients diagnosed with moderate-to-severe OSA (apnea-hypopnea index ≥ 15/hour) confirmed by PSG according to international criteria (American Academy of Sleep Medicine); patients aged between 18 and 65 years; patients able to comply with PSG procedures and the fascial mobilization protocol; patients with stable cardiac and respiratory conditions; patients without acute exacerbations or decompensated disease; patients who had not initiated continuous positive airway pressure, bilevel positive airway pressure, oral appliance, or surgical treatment, or who had discontinued such treatments at least 3 months prior; patients not using medications directly affecting sleep patterns or autonomic nervous system functions (e.g., sedatives, hypnotics, anticholinergics); and patients who agreed to participate by providing informed consent.

The exclusion criteria included patients with uncontrolled hypertension, advanced heart failure, severe arrhythmia, or severe asthma. Also excluded were patients with a history of coronary artery disease or cerebrovascular disease; chronic obstructive pulmonary disease or interstitial lung disease; epilepsy, Parkinson disease, or dementia; major depression or anxiety disorder; body mass index > 40 kg/m^2; those with a history of major surgery in the neck region (e.g., thyroidectomy, cervical disc surgery) or significant cervical structural pathology. Additional exclusions included patients with acute infections, skin lesions, or inflammatory disease in the neck/thorax region; patients continuing to use continuous positive airway pressure, bilevel positive airway pressure, or oral appliances; patients with narcolepsy, insomnia, or parasomnia; individuals consuming alcohol or sedative/hypnotic drugs; shift workers or individuals with irregular sleep schedules; pregnant or breastfeeding women; patients with cognitive impairment preventing adherence to the protocol; and patients with severe musculoskeletal limitations preventing mobilization in the neck/thorax region. Medication use was reviewed by the sleep medicine physicians during the clinical eligibility assessment. None of the participants included in the study were using antihypertensive medications, including beta-blockers or calcium-channel blockers, or other ongoing pharmacological treatments expected to directly affect heart rate or blood pressure at the time of enrollment.

During the recruitment period, 5 female patients were assessed for eligibility. Of these, 2 did not meet the age criterion, while 3 met other exclusion criteria. Consequently, no female participants were enrolled in the study.

The study was conducted in a sleep laboratory with an established PSG infrastructure. This research was conducted by a multidisciplinary team consisting of pulmonologists and physiotherapists to investigate physiotherapy-based interventions in individuals with OSA.

2.2. Sample size

Sample size estimation was based on Delaney et al.[19] An a priori power analysis (two independent groups, 2-tailed, $\alpha = .05$, $1 - \beta = 0.80$) was performed using post-intervention diastolic blood pressure values from a study examining the acute effects

of massage on autonomic function in healthy individuals (massage: 76 ± 2.5 mm Hg; control: 74 ± 2.5 mm Hg). Based on an expected mean difference of 2 mm Hg and a standard deviation of 2.5 mm Hg, the required sample size was calculated as at least 25 participants per group (50 participants in total). At the time the present study was planned, no previous study investigating the acute blood pressure responses to a comparable fascial mobilization intervention in individuals with OSA provided sufficient data for sample size estimation. Therefore, the study by Delaney et al, which investigated the acute physiological effects of a comparable manual intervention, was used as the reference for the sample size calculation. Calculations were performed using OpenEpi Version 3 software (an open-source statistical software package, available at: https://www.openepi.com/Menu/OE_Menu.htm).[20] The calculated sample size of 50 participants represented the minimum required sample rather than a predetermined recruitment ceiling. Recruitment was therefore not stopped when this minimum was reached; instead, all eligible patients who presented during the predefined recruitment period and met the study criteria were enrolled, resulting in a final sample of 95 participants. The participant recruitment, randomization, allocation, follow-up, and analysis processes are presented in Figure 1.

2.3. Randomization and masking

Participants were randomly assigned to either the intervention group (fascial mobilization) or the control group using a parallel-group design. Randomization was performed using the sealed envelope method, and allocation concealment was ensured using opaque envelopes.

This study was conducted as a single-blind RCT, with blinding limited to the outcome assessor. The physiotherapists who performed the intervention were not involved in the assessment process. All physiological measurements, including heart rate, systolic and diastolic blood pressure, and SpO_2, were recorded by a physiotherapist who was unaware of the participants' group assignments.

2.4. Outcomes

Demographic and clinical characteristics of all participants were recorded at baseline. Acute physiological parameters were measured under standardized conditions. Heart rate and blood pressure were assessed using the Omron M7 Intelli IT (HEM-7322T-E, Omron Healthcare Co.), and SpO_2 was measured using a Braun YK-81CEU pulse oximeter (Braun Healthcare). The outcome measures were selected based on a review of the literature, with preference given to noninvasive and readily assessable parameters that have been used in previous studies investigating the effects of manual therapies on the autonomic nervous system. In a systematic review examining the effects of manual therapies on autonomic nervous system activity, heart rate, blood pressure, and oxygen saturation were classified among the measures used to assess cardiovascular autonomic activity.[21] Additionally, acute

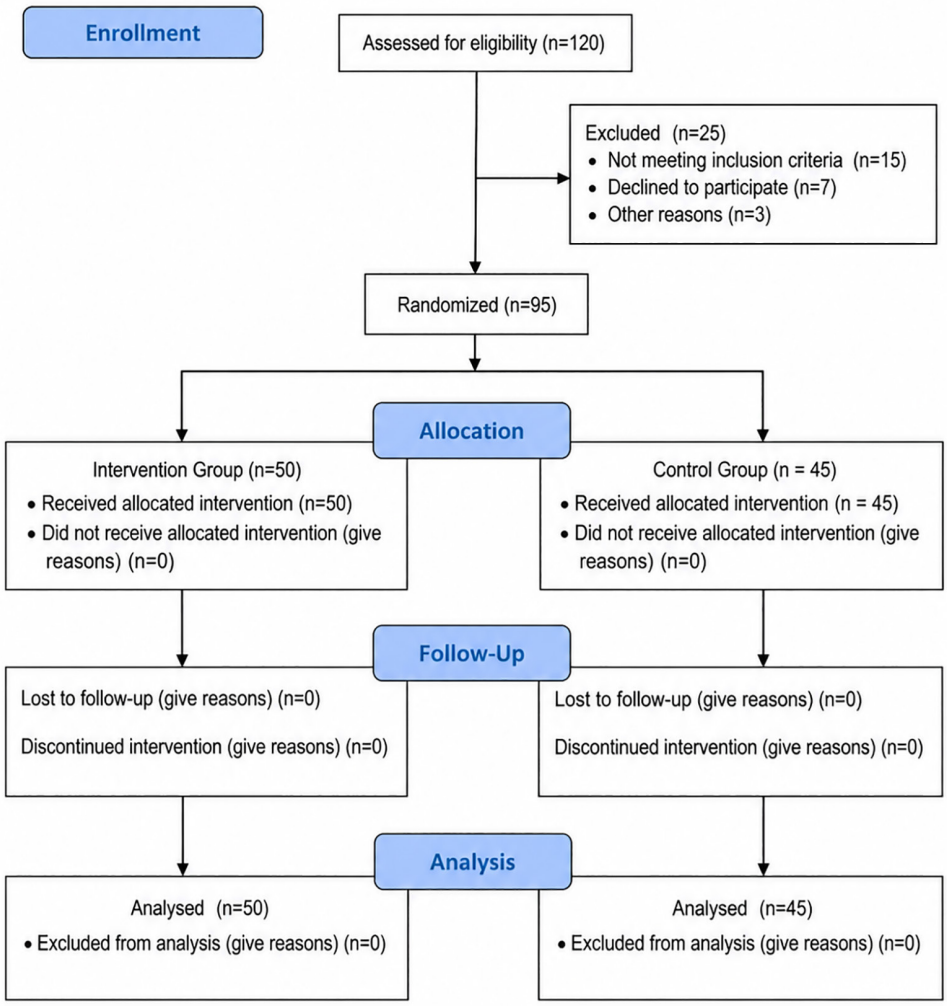


Figure 1. CONSORT flow diagram. CONSORT = consolidated standards of reporting trials, n = number of participants.

Table 1

Regions of mobilization and anatomical landmarks.

	Technique name	Definition	Anatomical rationale	Clinical rationale	Safety
1	Occipito-mastoid suture mobiliza-tion	Occipitomastoid suture mobilization has been described as a component of cranial suture traction approaches by Park et al.[23]	The occipitomastoid region is anatomically located near the skull base and jugular foramen, through which the vagus nerve exits the cranial cavity.	This region was selected because of its anatomical proximity to the jugular foramen and its inclusion in cranial mobilization approaches.	The weight of the head is used in a controlled manner, and no additional external force is applied.
2	Anterior Cervical Triangle	The anatomical boundaries and clinical significance of the anterior triangle of the neck have been described in detail by Shah and Cunha[24] highlighting its close relationship with the upper airway and cervical structures.	The carotid sheath is a critical region through which the vagus nerve and upper airway muscles pass.	This region was included because of its anatomical relationship with the ca-rotid sheath, upper airway, and surrounding cervical soft tissues.	Carotid artery compression is avoided, and mobilization is performed with minimal resistance.
3	Posterior Cervical Triangle	The posterior triangle of the cervical region is a neurovascularly rich area, and its topography and clinical significance have been detailed in the study by Kohan and Wirth.[25]	It is a region that contains the spinal accessory nerve, the brachial plexus, and lymphatic structures.	This region was included as a superficial cervical fascial region relevant to the multiregional mobilization protocol.	The application is restricted to superficial structures.
4	Hyoid Mobili-zation	Manual mobilization of the hyoid bone and surrounding soft tissues has been described by Lee and Yoon.[26]	The hyoid bone plays a central role in the stability of the tongue muscles and the upper airway.	Hyoid mobilization may influence local mobility and soft-tissue mechanics around the hyoid region.	No pressure is applied to the carotid arteries; mobilization is performed only on the hyoid cornua.
5	Sternum Mobiliza-tion	The anatomical structure of the sternum, including the manubrium, body, and xiphoid process, has been described by Altalib et al.[27]	The sternum regulates thoracic respira-tory movements by influencing costal mobility.	This region was included because of its anatomical contribution to anterior chest wall mechanics and its relationship with costal movement.	The application is limited to superficial pressure.
6	Diaphragm Mobiliza-tion	Manual mobilization techniques of the diaphragm and their effects on respira-tory functions have been described by Rocha[28] in addition, in Türkiye, Kömür[29] has further detailed this approach in terms of clinical application.	The diaphragm is the primary respiratory muscle and may exhibit functional limitations in OSA.	Diaphragm mobilization may promote expansion during the respiratory cycle and facilitate inspiratory muscle activation.	Mobilization is performed in synchronization with respiration, taking into account the patient's tolerance.
7	Trigonum Labbe	The anatomical location and clinical signif-icance of the Trigonum Labbe region, in the context of the relationships between the stomach and abdominal structures, have been addressed by Kahraman Yıldırım et al.[30]	It is the area where the gastric fundus is closest to the anterior abdominal wall.	This region was selected based on its anatom-ical relationship with upper abdominal visceral structures.	It is limited to light pressure, and no traumatic effect is exerted on the abdominal organs.
8	Radix Mes-enterii	The anatomy and clinical significance of this structure, where the mesentery of the small intestine attaches to the posterior abdominal wall, have been comprehensively described in classical anatomical sources.[31,32]	The radix mesenterii represents the attach-ment of the small-intestinal mesentery to the posterior abdominal wall and was included as an anatomical component of the prespecified multiregional fascial mobilization protocol.	This region was included because of its anatomical relationship with the mesenteric root and upper abdominal visceral structures.	It is performed with superficial palpation, avoiding excessive pressure.

myofascial interventions have been reported to induce short-term changes in these outcome measures.[22] However, these physiological parameters should not be considered direct or specific measures of vagal function. Rather, they represent clinically accessible physiological outcomes that may be influenced by autonomic regulation. Therefore, changes in these parameters were interpreted as short-term physiological and cardiovascular responses rather than as direct evidence of vagal activation.

Measurements were performed in a seated position after a standardized 5-minute rest period. At each measurement time point, heart rate and blood pressure were measured 3 times using the oscillometric device, with a 1-minute interval between consecutive measurements. The mean of the 3 measurements was used for statistical analysis. SpO_2 measurements were obtained sequentially using the pulse oximeter.

2.5. Intervention

The multiregional fascial mobilization protocol targeted 3 main anatomical regions: cervical, thoracic, and abdominal. Within these 3 regions, the intervention was applied to 8 discrete anatomical subsites, as detailed in Table 1. These subsites were selected based on anatomical and fascial considerations and included cervical sites along the anatomical course of the vagus nerve, as well as predefined thoracic and abdominal sites.[23–32] The total duration of the intervention was 16 minutes.

Although the intervention was anatomically oriented along the course of the vagus nerve, it did not involve direct neural manipulation or stimulation. Instead, a multiregional fascial mobilization approach was applied to examine its potential short-term physiological effects, which may involve peripheral fascial and mechanoreceptive pathways.

Table 2

Baseline demographic and clinical characteristics of the groups.

	Intervention group (n = 50)	Control group (n = 45)	Group comparison	
Variables	Mean ± SD Min–Max	Mean ± SD Min–Max	*t*	*P*
Age (yrs)	47.84 ± 9.79 (30–65)	48.4 ± 10.63 (28–65)	.250	.617
Height (m)	1.73 ± .07 (1.54–1.93)	1.74 ± .06 (1.64–1.9)	.285	.593
Weight (kg)	93.78 ± 15.46 (52–128)	92.06 ± 11.18 (74–116)	.305	.581
BMI (kg/m^2)	31.29 ± 5.44 (21.93–39.97)	30.26 ± 4.15 (21.61–39.21)	.397	.529
AHI	33.6 ± 23.41 (15–109)	33.16 ± 19.55 (15–84.9)	.165	.685

t: independent-samples *t*-test.

AHI = apnea-hypopnea index, BMI = body mass index, kg = kilogram; kg/m^2 = kilograms per square meter, m = meters, min = minimum, max = maximum, n = number of participants, SD = standard deviation.

The intervention consisted of superficial and deep fascial mobilization techniques. In superficial fascial mobilization, light pressure was applied using the fingertips or palms, and the skin was moved rhythmically until a sense of tissue gliding was achieved. In deep fascial mobilization, slow and sustained pressure was applied to areas of increased tissue tension. The pressure was progressively increased according to tissue resistance, followed by gentle stretching of the tissue.[33,34]

No direct electrophysiological or neural measurements were performed to assess vagal activity. Therefore, the observed changes should be interpreted as short-term physiological responses to the intervention rather than as evidence of vagal activation or modulation.

2.6. Control condition

Participants assigned to the control group did not receive fascial mobilization, manual therapy, or tactile intervention. They rested quietly in the supine position for a time period corresponding to the intervention session under the same standardized environmental conditions (a quiet, temperature-controlled room at 22–24°C). During this period, participants received no additional intervention beyond their routine clinical care and recommendations. Physiological measurements were obtained according to the same schedule and measurement protocol used for the intervention group.

2.7. Patient positioning and application standardization

Patients were placed in the supine position in a quiet, well-ventilated room with a controlled ambient temperature. Pillows were placed in the cervical and knee regions for relaxation and postural support. To ensure patient privacy, only the relevant anatomical area was exposed, whereas the remaining body regions were appropriately draped.[34] Each mobilization was applied for 120 seconds per region, resulting in a total intervention duration of 16 minutes. Patient comfort and safety were continuously monitored throughout the procedure, and short rest intervals were provided between applications when necessary. The fascial mobilization protocol was developed based on the clinical experience of the physiotherapists. All interventions were administered by experienced physiotherapists trained in fascial mobilization techniques. To ensure patient safety and clinical oversight, all applications were conducted under the supervision of 2 physicians specialized in sleep medicine. During the intervention, the participants were closely observed for any signs of discomfort, pain, or dizziness. No serious adverse events were reported during the study. The mobilized regions and anatomical reference points are presented in Table 1.

2.8. Statistical analysis

Continuous variables are summarized as mean ± standard deviation (min–max), while categorical variables are presented as frequencies and percentages (%). The distribution of the data was assessed using the Shapiro–Wilk test. Between-group comparisons were performed using the independent-samples *t*-test and, when appropriate, the chi-squared test. To examine changes over time and group × time interactions, a 2-way repeated-measures analysis of variance was applied. Because baseline heart rate values differed significantly between the groups, a mixed analysis of covariance (ANCOVA) was performed for heart rate. Only the 3 post-baseline measurements (immediately post-mobilization, 5 minutes post-mobilization, and 15 minutes post-mobilization) were included as repeated outcome measures, while pre-mobilization heart rate was included solely as a covariate to adjust for the baseline difference between groups. Thus, pre-mobilization heart rate was not included as a repeated outcome measure in the mixed ANCOVA. In cases where significant differences were detected, Bonferroni post hoc tests were conducted to determine the differences between the measurement time points. Statistical significance was set at $P < .05$. Effect sizes were reported using Partial Eta Squared (η^2).[35,36]

3. Results

3.1. Baseline characteristics

There were no statistically significant differences in the baseline demographic and clinical characteristics between the intervention and control groups ($P > .05$) (Table 2).

3.2. Between-group comparisons of physiological parameters

No significant differences were observed between the intervention and control groups in systolic blood pressure, diastolic blood pressure, and SpO_2 values at any measurement time point ($P > .05$). A significant difference was found only in the pre-mobilization heart rate values ($P = .047$) (Table 3).

3.3. Time effects and group × time interactions

After adjustment for baseline heart rate using mixed ANCOVA, a nominally significant time effect was observed for heart rate ($P = .046$, partial $\eta p^2 = .033$), whereas the group × time interaction was not significant ($P = .476$, partial $\eta p^2 = .008$).

Repeated-measures analysis of variance for the other physiological parameters revealed significant time effects for systolic blood pressure ($P < .001$, partial $\eta p^2 = .377$), diastolic blood pressure ($P < .001$, partial $\eta p^2 = .078$), and SpO_2 ($P = .010$, partial $\eta p^2 = .041$). However, no significant group × time interactions were observed for systolic blood pressure ($P = .475$, partial $\eta p^2 = .009$), diastolic blood pressure ($P = .192$, partial $\eta p^2 = .017$), or SpO_2 ($P = .615$, partial $\eta p^2 = .006$) (Table 4).

3.4. Within-group changes over time

Within-group analyses demonstrated a significant time effect for heart rate in the intervention group ($P = .026$), but not in the control group ($P = .713$). For heart rate, pairwise comparisons involving the pre-mobilization measurement (P1–P3) were not performed because pre-mobilization heart rate was used as a covariate in the adjusted analysis; only comparisons among the 3 post-baseline time points (P4–P6) are presented in Table 5. Within-group changes in systolic blood pressure, diastolic blood pressure, and SpO_2 are also presented in Table 5. These within-group analyses are descriptive, whereas the primary between-group longitudinal analysis for heart rate was the mixed ANCOVA adjusted for baseline heart rate, as presented in Table 4.

Changes over time in SpO_2, heart rate, systolic blood pressure, and diastolic blood pressure are presented in Figures 2–5, respectively.

4. Discussion

This RCT investigated the acute effects of fascial mobilization applied along the anatomical course of the vagus nerve in patients with OSA. Although the baseline characteristics of the intervention and control groups were generally comparable, baseline heart rate differed significantly between the groups. Therefore, baseline heart rate was controlled for as a covariate. In the adjusted analysis, although the main effect of time on heart rate was significant, the group × time interaction was not significant. Similarly, no significant group × time interactions were observed for systolic and diastolic blood pressure or SpO_2. These findings indicate that changes in the evaluated physiological parameters occurred over time; however, these changes appear to reflect time-dependent responses observed in both groups rather than an effect specifically attributable to fascial mobilization.

Sleep-disordered breathing is characterized by recurrent apneas and hypopneas associated with hypoxemia, hypercapnia, sleep fragmentation, and altered autonomic regulation.[37] These disturbances contribute to the cardiovascular

Table 3

Comparison of hemodynamic and SpO_2 values before and after mobilization between groups.

		Intervention Group (n = 50)	Control Group (n = 45)		
Measurement time points	**Variables**	**Mean ± SD Min–Max**	**Mean ± SD Min–Max**	***t***	***P***
Pre- mobilization	Heart Rate (bpm)	81.74 ± 11.39 (60–106)	77.06 ± 11.17 (57–102)	2.00	.047*
	SBP (mm Hg)	126.96 ± 14.3 (100–162)	125.75 ± 13.5 (107–166)	.420	.675
	DBP (mm Hg)	79.68 ± 11.33 (61–111)	76.3 ± 9.95 (57–100)	1.52	.131
	SpO_2 (%)	94.98 ± 1.77 (89–98)	95.4 ± 1.99 (91–99)	–1.08	.281
Post-mobilization	Heart Rate (bpm)	74.14 ± 10.17 (53–97)	73.8 ± 11.46 (53–103)	.153	.879
	SBP (mm Hg)	119.3 ± 12.04 (98–151)	119.8 ± 13.48 (103–168)	–.191	.849
	DBP (mm Hg)	77.46 ± 9.98 (61–104)	75.04 ± 9.40 (60–108)	1.210	.229
	SpO_2 (%)	94.24 ± 2 (89–98)	94.8 ± 2.35 (88–99)	–1.449	.151
5 min post-mobilization	Heart Rate (bpm)	73.68 ± 8.82 (57–90)	74.8 ± 11.32 (57–99)	–.540	.590
	SBP (mm Hg)	119.7 ± 13.48 (97–154)	118.28 ± 14.5 (96–163)	.491	.624
	DBP (mm Hg)	76.96 ± 10.33 (59–107)	75.42 ± 10.05 (58–109)	.734	.465
	SpO_2 (%)	94.48 ± 1.96 (91–99)	94.71 ± 2.52 (89–99)	–.500	.618
15 min post-mobilization	Heart Rate (bpm)	74.3 ± 9.12 (58–96)	74.73 ± 11.5 (52–98)	–.204	.839
	SBP (mm Hg)	119.34 ± 12.65 (99–157)	118.75 ± 13.22 (100–168)	.220	.826
	DBP (mm Hg)	76.56 ± 10.44 (59–111)	75.1 ± 9.95 (60–110)	.690	.492
	SpO_2 (%)	94.74 ± 1.91 (90–99)	94.8 ± 2.21 (90–99)	–.351	.726

mm Hg means millimeters of mercury; *t*: independent-samples *t*-test.

bpm = beats per minute, DBP = diastolic blood pressure, SBP = systolic blood pressure, SD = standard deviation, SpO_2 = peripheral oxygen saturation.

*$P < .05$ was considered statistically significant.

Table 4

Time and Group × Time interaction effects on hemodynamic parameters and SpO_2.

	Time		Time*Group	
	P	Partial Eta square (time)	P	Partial Eta square (Time*Group)
Heart Rate (bpm)	.046*	.033	.476	.008
SBP (mm Hg)	<.001	.377	.475	.009
DBP (mm Hg)	<.001	.078	.192	.017
SpO_2 (%)	.010*	.041	.615	.006

The Time effect indicates whether there are significant differences across measurement time points, whereas the Time × Group effect reflects whether these differences vary between groups. For heart rate, mixed ANCOVA was performed using only the 3 post-baseline measurements (immediately post-mobilization, 5 minutes post-mobilization, and 15 minutes post-mobilization) as repeated outcome measures, with pre-mobilization heart rate included solely as a covariate to adjust for the baseline difference between groups. Thus, the pre-mobilization heart rate was not included as a repeated outcome measure in the mixed ANCOVA.

ANCOVA = analysis of covariance, DBP = diastolic blood pressure, SBP = systolic blood pressure, SpO_2 = peripheral oxygen saturation.

Table 5

Within-group comparisons of hemodynamic and SpO_2 parameters across time points.

		Time							
		P	Partial Eta square	P^1	P^2	P^3	P^4	P^5	P^6
Intervention Group (n = 50)	HR (bpm)	.026*	.073	-	-	-	1.000	1.000	.807
	SBP (mm Hg)	< .001	.367	< .001	< .001	< .001	-	-	-
	DBP (mm Hg)	.001†	.123	.070	.023	.003†	-	-	-
	SpO_2 (%)	.039†	.055	.037†	.511	-	-	.375	-
Control Group (n = 45)	HR (bpm)	.713	.008	-	-	-	.367	.506	1.000
	SBP (mm Hg)	< .001	.403	< .001	< .001	< .001	-	-	-
	DBP (mm Hg)	.151	.041	.404	-	.656	-	-	-
	SpO_2 (%)	.152	.039	.780	.229	.448	-	-	-

1 = pre-mobilization, 2 = post-mobilization, 3 = 5 minutes post-mobilization, 4 = 15 minutes post-mobilization. Time indicates whether there were significant within-group differences across time points. P^1, P^2, P^3, etc, represent the specific pairwise comparisons between measurement times (P^1:1 × 2, P^2:1 × 3, P^3:1 × 4, P^4:2 × 3, P^5:2 × 4, P^6:3 × 4). Pairwise *P* values (P^1–P^6) were adjusted using the Bonferroni correction for multiple comparisons. For heart rate, within-group analyses are presented to describe changes over time within each group. The primary between-group longitudinal analysis for heart rate was performed using mixed ANCOVA adjusted for baseline heart rate (Table 4). For heart rate (bpm), P^1–P^3 pairwise comparisons were not performed because the pre-mobilization heart rate value was used as a covariate in the analysis.

ANCOVA = analysis of covariance, DBP = diastolic blood pressure, HR = heart rate, n = numner of participants, SBP = systolic blood pressure, SpO_2 = peripheral oxygen saturation.

burden associated with OSA and provide a clinical rationale for examining short-term cardiovascular responses to non-pharmacological interventions.[38,39] In the present study, however, the temporal patterns of heart rate, blood pressure, and SpO_2 did not differ significantly between fascial mobilization and the resting control condition. Thus, the findings do not provide evidence of an intervention-specific physiological effect. Moreover, because direct measures of autonomic function were not included, the present outcomes cannot determine whether either intervention altered autonomic or vagal activity.

Previous studies have suggested that manual therapy and myofascial techniques may influence autonomic responses and heart rate variability (HRV)-related indices of autonomic regulation.[40,41] Roura et al[21] emphasized that manual therapy interventions affect both sympathetic and parasympathetic activities, although the findings remain heterogeneous. Jupin et al[42] demonstrated that spinal manual therapy may induce short-term neurophysiological and autonomic changes, whereas de Sousa et al[43] reported acute improvements in cardiac autonomic control. In line with this, Alanazi et al[44] observed short-term physiological responses to manual therapy in heart rate, blood pressure, and respiratory parameters, whereas Stępnik et al[45] showed that osteopathic techniques can influence autonomic and spirometric outcomes, though inconsistently. Navarro-Rico et al[41] reported short-term changes in HRV parameters following visceral manual therapy, suggesting possible modulation of autonomic activity. Collectively, these findings indicate that manual therapeutic interventions may be associated with short-term autonomic and cardiovascular responses. However, since the time-dependent changes observed in heart rate did not differ between the groups in the present study, these changes should not be interpreted as an autonomic response specific to fascial mobilization. Furthermore, as no direct measures of autonomic activity were performed, these findings cannot be considered evidence of vagal activation.

Persistent sympathetic overactivity is recognized as a major determinant of cardiovascular risk in OSA and contributes to hypertension and arrhythmias.[46,47] Although heart rate changed significantly over time in the present study, the absence of a significant group × time interaction after adjustment for baseline heart rate indicates that this change was not specific to fascial mobilization. Significant time effects were also observed for systolic blood pressure, diastolic blood pressure, and SpO_2, with effect sizes varying across outcomes. Notably, the time effect was larger for systolic blood pressure (partial $\eta p^2 = .377$) than for heart rate (partial $\eta p^2 = .033$), diastolic blood pressure (partial $\eta p^2 = .078$), and SpO_2 (partial $\eta p^2 = .041$). However, none of these outcomes showed a significant group × time interaction. Therefore, the magnitude of the observed time effects should be distinguished from evidence of an intervention-specific effect. Because the control group rested quietly for the corresponding period without receiving mobilization or tactile intervention, the lack of significant group × time interactions indicates that the temporal physiological responses observed following fascial mobilization were not distinguishable from those occurring

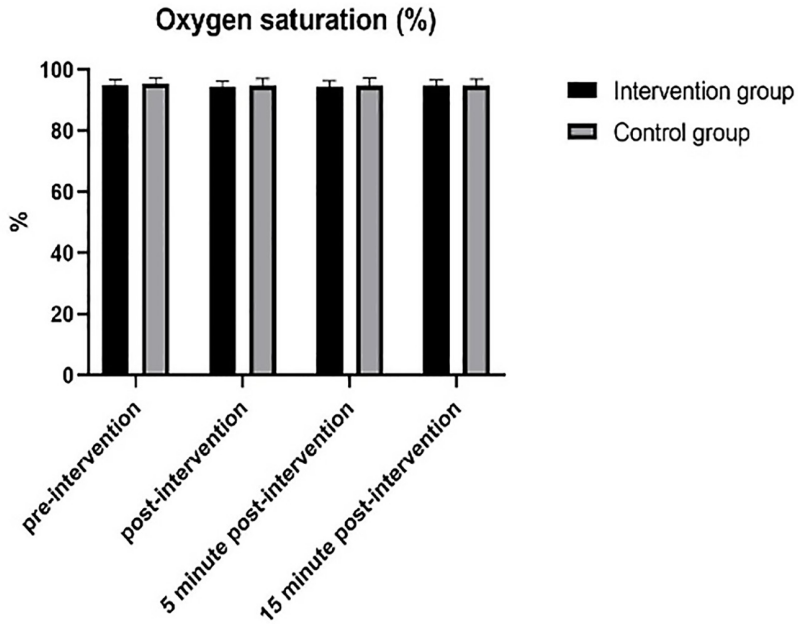


Figure 2. Changes in oxygen saturation over time in the intervention and control groups.

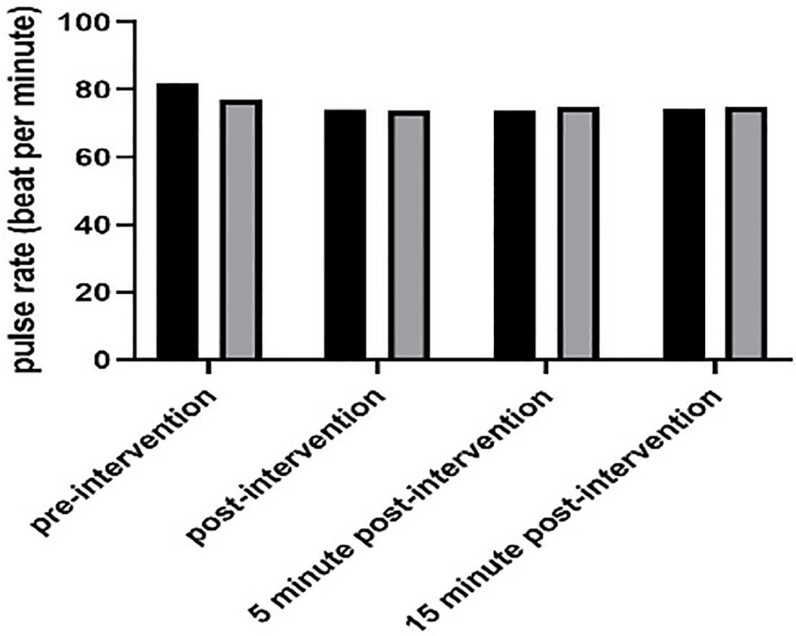


Figure 3. Changes in heart rate over time in the intervention and control groups.

during the resting control condition. Accordingly, the present findings do not provide evidence of a specific acute physiological effect of multiregional fascial mobilization for the outcomes assessed.

To our knowledge, this is one of the first clinical studies to investigate the short-term physiological responses to fascial mobilization applied along the anatomical course of the vagus nerve in patients with OSA. Although this approach is noninvasive and relatively simple to administer, the present findings did not demonstrate a specific effect of fascial mobilization compared with the resting control condition on the physiological parameters assessed. Therefore, further controlled studies incorporating direct measures of autonomic function are needed before conclusions can be drawn regarding its potential role in autonomic modulation or its clinical utility in OSA management.

5. Limitations

This study has several limitations. First, only acute effects were evaluated, and long-term outcomes were not investigated. In addition, only basic physiological parameters were assessed, and no direct electrophysiological or autonomic measurements were performed to assess vagal activity. Therefore, the observed physiological changes should not be interpreted as direct evidence of vagal activation or modulation. Future studies should incorporate more advanced measures, such as HRV and baroreflex sensitivity, to provide a more comprehensive assessment of autonomic responses.

Second, although female sex was not an exclusion criterion, no female participants were enrolled. During the recruitment period, 5 female patients were assessed for eligibility; 2 did not meet the age criterion, and 3 met other exclusion criteria.

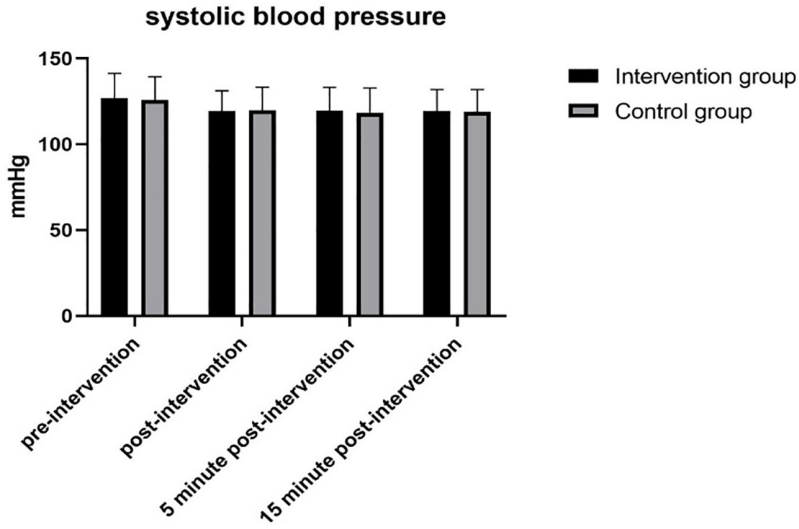


Figure 4. Changes in systolic blood pressure over time in the intervention and control groups.

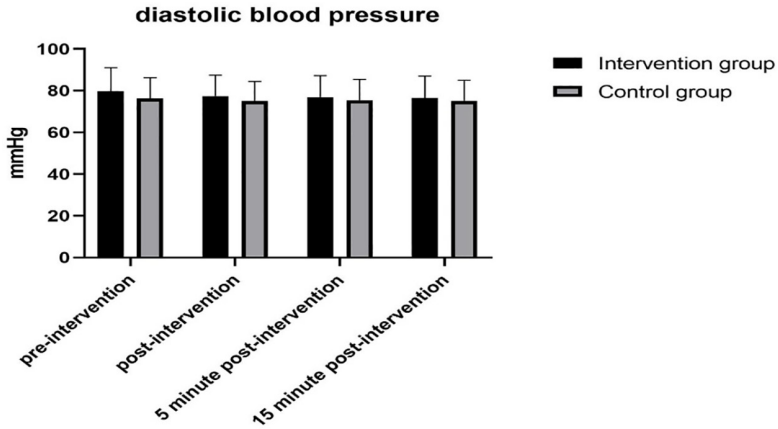


Figure 5. Changes in diastolic blood pressure over time in the intervention and control groups.

Consequently, the study sample consisted exclusively of male participants. Given potential sex-related differences in autonomic and cardiovascular responses, this limits the generalizability of the findings to female patients. Future studies should therefore aim to include participants of both sexes.

Third, the a priori sample size estimation was based on indirect evidence. At the time the study was designed, directly applicable preliminary data on the acute physiological effects of multiregional fascial mobilization in patients with OSA were unavailable. Therefore, the effect-size estimate was derived from a previous massage-based intervention conducted in healthy individuals. Because both the population and the intervention differed from those examined in the present trial, the effect estimate cannot be considered directly representative of the expected treatment effect in patients with OSA undergoing multiregional fascial mobilization. This mismatch introduces uncertainty regarding the adequacy of the planned sample size for detecting between-group effects. Furthermore, the calculated sample size of 50 participants represented the minimum required sample rather than a predetermined recruitment ceiling. Recruitment continued throughout the predefined recruitment period, during which all eligible patients who met the study criteria were enrolled, resulting in a final sample of 95 participants. Although the larger sample increased statistical precision, it may also have increased the ability to detect small temporal differences as nominally statistically significant. This should be considered when interpreting the observed time effects, particularly in the absence of significant group × time interactions.

Fourth, no family-wise multiplicity correction was applied across the 4 physiological outcome families. Therefore, nominally significant time effects, particularly those close to the conventional significance threshold, should be interpreted cautiously and considered exploratory rather than definitive.

Finally, the control condition consisted of time-matched quiet rest without manual or tactile intervention. Although this design allowed the physiological responses following fascial mobilization to be compared with those occurring during rest under similar environmental conditions, participants could not be blinded to whether they received manual treatment. Therefore, expectancy and other nonspecific contextual effects cannot be completely excluded. Outcome assessment was nevertheless performed by an assessor blinded to group allocation.

6. Conclusion

This study showed time-dependent changes in heart rate in individuals with OSA. However, after adjustment for baseline heart rate, the pattern of heart rate change did not differ significantly between the fascial mobilization and resting control groups. Similarly, no significant group × time interactions were observed for blood pressure or oxygen saturation. Therefore, the present findings do not demonstrate a specific acute effect of multiregional fascial mobilization on the physiological parameters

assessed. The temporal changes observed in both groups may reflect nonspecific effects associated with rest, relaxation, habituation to the measurement setting, or other time-related factors.

No adverse events were observed during the intervention, supporting the short-term tolerability of the procedure. However, because direct measures of autonomic function were not assessed, no conclusions can be drawn regarding vagal activation or autonomic modulation. Future studies should investigate the potential physiological and clinical effects of this approach using direct autonomic measures, including HRV and baroreflex sensitivity, together with sleep-related outcomes and longer follow-up periods.

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